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Semantic Mapping of Scholarly Publisher Catalogues and Scientific Context: A Longitudinal Study Using OpenAlex and Research Horizon

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Abstract

Scholarly publisher catalogues provide a long-term record of how knowledge is organized, developed and communicated, but their heterogeneous text and uneven metadata make them difficult to analyse systematically over time. This study presents a semantic framework for representing publisher catalogues through explicit thematic, temporal, hierarchical and provenance information, while also relating internal editorial trajectories to external scientific activity. The framework is demonstrated using the 1978–2026 catalogue of Editorial ABYA-YALA. Semantic topic modelling is combined with documented expert curation to reconstruct the catalogue's thematic structure and trace its evolution over nearly five decades. Selected editorial lines are then compared with scientific activity indexed in OpenAlex and with screened emerging research fronts from Research Horizon. The results reveal a stable long-term thematic core alongside substantial recent recomposition and several lines showing renewed activity. They also show that historical prominence within the catalogue, current scientific activity, and proximity to emerging research fronts do not necessarily coincide. By preserving both algorithmic and curated states, the framework makes expert intervention and cross-source interpretation traceable.

Keywords: knowledge representation; semantic topic modeling; scholarly publishers; longitudinal analysis; OpenAlex; research fronts

1. Introduction

Scholarly publisher catalogues provide a long-term record of how thematic priorities develop, persist and change. Yet they are difficult to analyse systematically because they combine heterogeneous natural-language descriptions, incomplete and uneven metadata, long temporal spans, editorial provenance and links to external scientific knowledge infrastructures. An adequate representation must therefore preserve semantic, temporal, hierarchical and provenance information.

This challenge is particularly relevant in fields where books remain an important form of scholarly communication and journal-based infrastructures provide only a partial view of research activity [1–3]. Research on scholarly books and publishers has consequently developed several complementary forms of evaluation. Publisher surveys and rankings address perceived prestige [4]; citation analyses examine scholarly impact, while library

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holdings provide an alternative indication of use or circulation [5,6]; and bibliographic infrastructures such as the Book Citation Index provide partial coverage of book-based scholarship [7]. These approaches assess impact, prestige, visibility and circulation, but say less about the catalogue itself as a changing thematic system.

Scientometric research provides methods that could help address this thematic and longitudinal problem. Co-citation and science-mapping methods recover relationships among scientific documents and specialties [8–10]. Topic models and semantic clustering represent large corpora through latent or embedding-based thematic structures [11–13]. Longitudinal methods trace changes in topic distributions, the movement of venues through topic space and paths of scientific development [14,15], while research-front studies identify emerging clusters and characterize recent scientific activity [16–18]. Together, these methods offer tools for identifying thematic organization and tracking change. Their usual empirical objects, however, are scientific documents, venues, specialties or fields rather than the long-term thematic structure of a publisher catalogue.

The resulting gap lies in bringing these perspectives together. A publisher catalogue must be represented as an evolving thematic structure while preserving its semantic organization, temporal development, hierarchical context and editorial provenance. Selected thematic lines must also be related to external scientific activity without conflating the different forms of evidence.

This study develops such a framework and demonstrates it with the 1978–2026 catalogue of Editorial ABYA-YALA, an Ecuadorian scholarly publisher. The catalogue's nearly five-decade span and breadth across social-science, humanities, educational, environmental, religious and technical subjects make it a useful case for examining long-running editorial lines and recent recomposition. The empirical objective is to reconstruct its thematic development and situate selected lines in relation to topic-country indexed activity in OpenAlex and screened emerging research fronts from Research Horizon.

The framework makes four contributions in this proof-of-concept:

- First, it derives explicit topics from heterogeneous catalogue text through embedding-based clustering and documented expert curation, retaining the initial partition, curated state and document-level changes as separate stages.
- Second, it organizes the topics through hierarchical context and longitudinal attributes.
- Third, it formalizes a case-specific renewal decision as a two-stage procedure that separates quantitative growth from trajectory qualification. The frozen case makes the rule executable, while future replications can pre-specify and validate its thresholds independently.
- Fourth, it relates selected topics to OpenAlex topic-country indexed activity and a corpus-wide documentary review of screened Research Horizon candidates.

The architecture and audit trail are intended to be reusable across cases, whereas the numerical thresholds, focal-line selection, country set, hierarchy cut, mapping roles and Research Horizon screening design require case-specific specification and validation.

The remainder of the article proceeds from conceptual framing to empirical demonstration. Section 2 develops the related literature and analytical framework and states the research questions. Section 3 describes the data and methods, with emphasis on reproducibility and on the separation between algorithmic and expert-assisted stages. Section 4 reports the longitudinal proof-of-concept. Section 5 interprets the findings, states their methodological implications and discusses limitations. Section 6 concludes and identifies concrete directions for replication and external validation.

2. Related work and analytical framework

2.1. Scholarly books, publishers and multidimensional evaluation

Research on scholarly communication shows that, in many social-science and humanities fields, journal articles alone do not capture the full publication portfolio. Books and chapters retain important roles in several fields, publication languages are more diverse and locally or nationally oriented channels may coexist with international journals [1–3,19]. These patterns have direct consequences for evaluation. Systems designed around indexed journal output can underrepresent types of scholarship for which books are an important communication form, while book-specific evaluation confronts uneven metadata, partial citation coverage and heterogeneity among publishers [20,21].

One response has been to evaluate the publisher as an intermediary. Publisher prestige surveys have generated structured assessments in the social sciences and humanities [4]. Bibliometric experiments have examined publisher ranking from citation data [22], while other work has analyzed citations to books, publisher coverage and library holdings as alternative evidence of impact or use [5–7]. These approaches operationalize different constructs: prestige captures perceived standing, citations scholarly uptake, holdings library circulation, database inclusion indexed visibility and publication counts output volume. The Leiden Manifesto's broader emphasis on contextualized indicators and heterogeneous forms of research performance is therefore directly relevant [23].

Bibliodiversity reinforces this methodological point. Recent work treats linguistic, geographic, disciplinary, publisher, business-model and infrastructural diversity as connected but separable dimensions [24]. We apply the same measurement principle to thematic diversity, persistence, indexed scientific activity and retention of screened research-front candidates. The internal catalogue and the two external sources are analyzed separately and then compared.

This choice also changes the object of analysis. The central object becomes the catalogue as a longitudinal thematic system. Its internal structure can then be related to external scientific classifications, while causal influence and direct citation impact remain separate empirical questions that require publisher-to-publication linkage data.

2.2. Longitudinal thematic mapping and human interpretation

Science mapping has a long tradition of representing intellectual structures from relations among documents, citations, terms, journals or other scientific entities [8,9,25–27]. Topic modeling introduced another route by representing documents through distributions over latent themes [11], while dynamic topic models explicitly incorporated temporal change [12]. More recent embedding-based approaches make it possible to group semantically similar documents without requiring a generative word model. BERTopic, for example, combines document embeddings, dimensionality reduction, density-based clustering and class-based TF-IDF representations [13].

The methodological benefit of these tools is accompanied by an interpretability requirement. Topic solutions need substantive interpretation because apparently coherent numerical structures can still combine heterogeneous documents under a convenient label. Human evaluations of topic models have therefore remained important even when statistical coherence measures are available [28,29]. Longitudinal applications add another complication: a stable label can conceal changes in composition, while a large cluster can reflect a documentary series or production mechanism instead of a substantive research trajectory.

Work on topic-space trajectories is particularly relevant because it treats temporal movement and interpretability as core analytical problems rather than visual afterthoughts. Schaefermeier et al. [14] developed topic-space trajectories to follow publication venues

over time and emphasized interpretable representation. Kim et al. [15] integrated topic information with path extraction to explore scientific trajectories at scale. Our object differs from both cases: a publisher catalogue contains books produced under an editorial history, not papers grouped by scientific venue or citation path. Nevertheless, the trajectory perspective provides a useful foundation for analyzing continuity, recomposition and thematic movement across a long time window.

For an editorial catalogue, we use a hybrid design that separates the initial algorithmic partition from expert curation. Clustering provides a reproducible grouping from frozen model-ready inputs, while expert review can identify editorial structures visible in documentary records but not necessarily in semantic density. Storing both states allows the intervention and its effects to be quantified.

2.3. External scholarly context and research-front evidence

Internal thematic structure answers one part of the evaluation problem. A historically persistent catalogue line may correspond to a field with substantial present-day scientific activity, or it may occupy a narrower or more regionally specific niche. Open bibliographic infrastructures allow these relationships to be examined at scale. OpenAlex is an open scholarly graph and structured knowledge infrastructure containing entities such as works, authors, venues, institutions and topics [30]. Its coverage is broader than that of several conventional databases in settings relevant to Latin American social sciences and humanities, including Spanish-language output [31]. Coverage nevertheless remains source-dependent: language metadata can overestimate English-language representation [32], and broader bibliographic comparisons show database-specific differences [33]. In this study, OpenAlex supplies structured external context for the mapped topics through topic-country indexed scientific activity.

Research-front analysis poses a different problem. The classical literature has used co-citation, bibliographic coupling, direct citation, clustering and temporal signals to identify specialties and emerging areas [16,17]. Emergence involves more than growth: novelty, persistence, coherence and community formation can matter, and apparently fast-growing signals can be transient [18]. The candidate research fronts used here come from Clarivate's Research Horizon Navigator, which identifies emerging topics as co-citation clusters formed by a core of highly cited papers and the recent papers citing that core; these emerging topics are treated here as candidate research fronts in the sense of the literature above [16,17]. The exported topic universe serves as a fixed candidate source. Expert-assisted screening based on metadata similarity first selects candidate topics for each editorial line; corpus-wide documentary review of the retrieved titles, abstracts, keywords, subject information, affiliations and representative papers then classifies each screened candidate as a principal, complementary or specific-subline relation, or as reserve or excluded. The primary result is the number and share of screened candidates retained after documentary review, together with the composition of the retained relations. Reporting the screened denominator makes candidate-set size explicit in every line-level comparison.

The two external sources therefore answer different questions. OpenAlex characterizes current indexed scientific activity around the mapped topics across a pre-specified set of countries. Research Horizon identifies emerging research fronts, and the documentary review establishes which of the screened fronts are substantively related to each editorial line. Both sources are linked to the internal thematic map, and their outputs are read on their own scales.

2.4. Research questions and analytical framework

These literatures lead to four empirical research questions. The first characterizes the mapped stable thematic core, the second quantifies the effect of documented curation and the last two connect selected editorial lines to external evidence.

- **RQ1.** What longitudinal thematic structure characterizes the mapped stable semantic core of the ABYA-YALA catalogue in terms of concentration, diversity, temporal persistence, compositional change and recent renewal?
- **RQ2.** How materially does documented expert curation alter the initial algorithmic semantic partition and downstream diversity measures?
- **RQ3.** How does indexed scientific activity around nine focal editorial lines vary within the pre-specified ten-country OpenAlex comparison?
- **RQ4.** Within the screened Research Horizon candidate set, how does candidate retention after documentary review vary across focal lines, and where does it diverge from internal accumulation and OpenAlex indexed activity?

Figure 1 summarizes the architecture. The internal layer combines semantic structure, longitudinal indicators and substantive-renewal classification; OpenAlex indexed activity and screened research-front retention are evaluated independently.

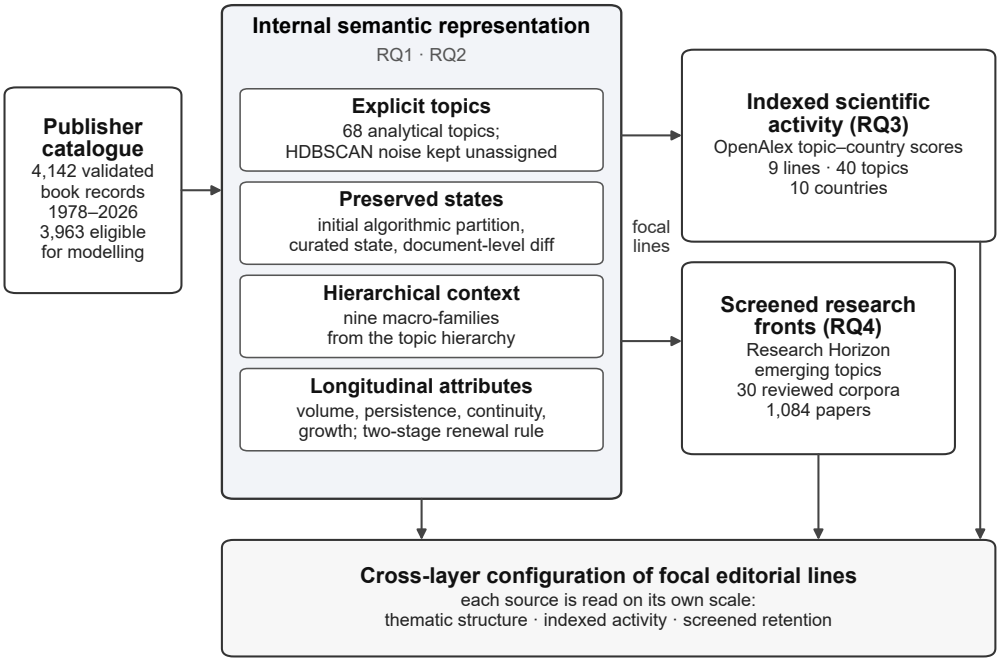


Figure 1. Analytical architecture of the framework: the catalogue feeds the internal semantic representation, selected lines are related to OpenAlex activity and screened Research Horizon fronts, and the three sources are compared on their own scales.

Table 1 makes the research-question traceability explicit. This mapping is defined before the empirical results so that the evidence used to answer each question is clear.

3. Data and methods

The study in this paper is a longitudinal methodological proof-of-concept. It reconstructs a publisher's thematic trajectory from frozen model-ready inputs and relates it to indexed activity and screened research-front candidates. The design has four stages:

1. a validated publisher catalogue;

Table 1. Research questions, their primary evidence, methodological operations and results sections

RQ	Primary evidence	Methodological operation	Main results location
RQ1	Curated catalogue, longitudinal topic series	Semantic mapping; Hill diversity, concentration, persistence, continuity, compositional distance and reference renewal classification	Sections 4.2–4.4
RQ2	Initial algorithmic and curated topic states	Document-level curation diff; pre/post diversity and family comparison; hierarchy sensitivity	Sections 4.1–4.2
RQ3	40 unique OpenAlex topics mapped to nine editorial lines	Topic-country activity model; line-country aggregation; bounded territorial and weight sensitivity	Section 4.5
RQ4	11,325 Research Horizon topic IDs, 73 screened associations, 30 candidate corpora	Expert-assisted screening; corpus-wide documentary review; retained-candidate fraction and role composition	Sections 4.6–4.7

2. an algorithmic topic partition followed by documented expert curation;
3. longitudinal measures and thematic-renewal classification; and
4. mappings to OpenAlex and Research Horizon.

Figure 2 shows the corpus-to-map sequence, and Table 2 summarizes the evidence design. The internal layer combines topic membership, hierarchical family context, longitudinal attributes and a record of curation. OpenAlex activity and Research Horizon review outcomes are then attached to selected topics without rescaling them to a common measure.

Table 2. Evidence sources, the coverage used in this study and their analytical role

Source	Coverage used in this study	Analytical role
Unified ABYA-YALA book catalogue	4,142 validated book records, 1978–2026; cut-off 22 July 2026	Internal longitudinal evidence; title/abstract semantic representation; book as unit of analysis
Reproducibility package	3,963 semantically eligible records; SPECTER2 embeddings, anchored UMAP arrays, initial algorithmic and curated topic states	Algorithmic topic discovery, curation audit, hierarchy, annual topic series, reproducibility
OpenAlex	40 unique topics, 46 editorial-line mappings, 10 pre-specified countries, 2015–2025 window	Comparative indexed scientific activity: accumulated/recent activity, world share, international collaboration, growth
Research Horizon	11,325 topic IDs (11,232 unique names); 73 candidate associations; 30 Priority-1 datasets; 1,084 papers	Candidate research-front screening and corpus-wide documentary review

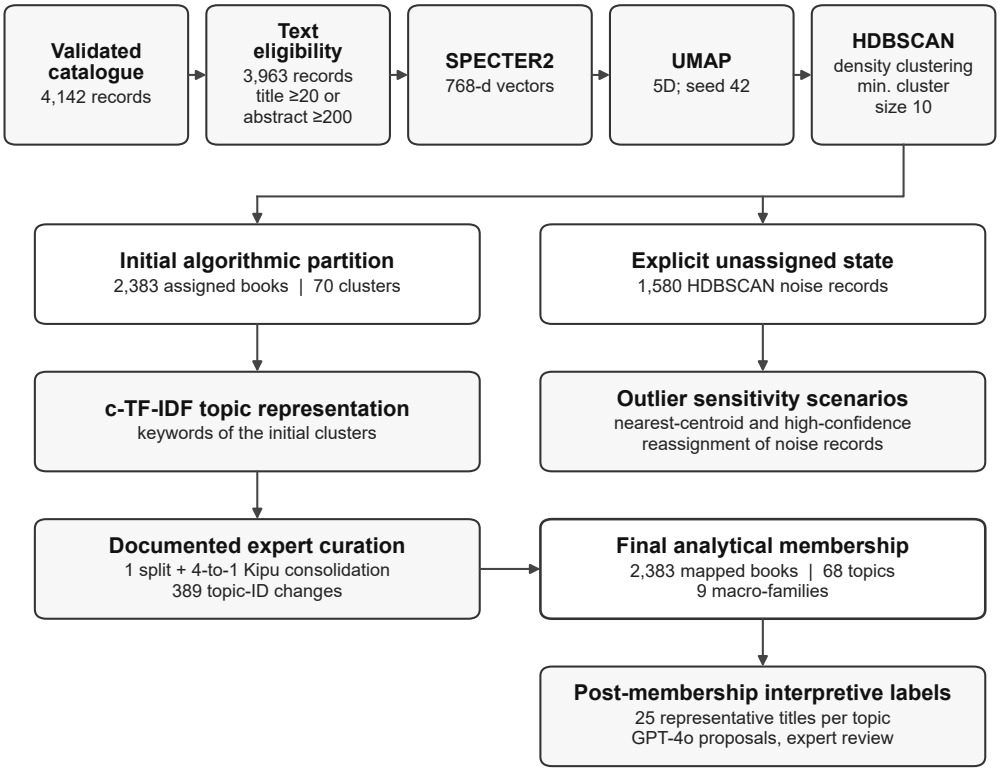


Figure 2. Semantic-mapping workflow from the validated catalogue to the final analytical topics.

3.1. Catalogue, unit of analysis and text eligibility

The source is the unified catalogue of books published by Editorial ABYA-YALA. The catalogue export lists one row per author–book combination; collapsing these rows to one record per book yielded 4,142 validated book records, which form the documentary universe of the study.

The unit of analysis throughout the internal layer is the validated book record. The year variable is the editorial year available in the catalogue. A record entered the semantic analysis if it satisfied at least one of two text-availability conditions: a title of at least 20 characters *or* an abstract or summary of at least 200 characters. This canonical OR rule yielded 3,963 semantically eligible books and left 179 validated catalogue books outside the embedding-based analysis. For each eligible record, the model-ready input is the English-translated title followed, when available, by the English-translated abstract or summary—the reproducibility package preserves these exact strings together with the original titles. Translation occurred upstream in the analytical platform and is treated here as a fixed preprocessing input; it is not re-estimated in the reproducibility package. The semantic analysis therefore reproduces the model from frozen text inputs rather than auditing the translation procedure itself.

The temporal coverage begins in 1978, but the early years are sparse and the source contains no record for 1979. We therefore distinguish a 1978–1986 prelude from the effective long-run series beginning in 1987. The main interpretive periodization is 1987–1998 (foundation), 1999–2008 (diversified plateau), 2009–2014 (maximum expansion), 2015–2021 (contraction and transition) and 2022–2026 (recent recomposition). The 2025 and 2026 records reflect the latest catalogue update and are incomplete relative to earlier years; recent descriptive comparisons therefore give priority to the complete 2022–2024 observations. The Kipu documentary series receives a separate chronological treatment because its

catalogue year tracks editorial registration and does not necessarily reproduce the original date of each underlying periodical item.

3.2. Semantic representation and initial algorithmic topic discovery

Documents were represented with SPECTER2, producing a 768-dimensional vector for each semantically eligible book. SPECTER2 was developed for scientific document representation and evaluated across heterogeneous scientific tasks and fields [34]. Its disciplinary breadth provides a practical reason for using it with a catalogue that spans social sciences, humanities, education, environmental studies and technical fields. Its transfer from scientific documents to scholarly monographs is treated as a specific source of model uncertainty and is evaluated within the study's limitations.

The initial algorithmic topic model follows the BERTopic architecture [13]. SPECTER2 embeddings were projected to five dimensions with UMAP [35] and then clustered with HDBSCAN [36]. BERTopic used the resulting cluster memberships to construct class-based TF-IDF topic representations. A separate two-dimensional UMAP projection was retained only for visualization; it did not determine cluster membership. Table 3 lists the fixed software versions and primary parameters.

Table 3. Frozen topic-model configuration, software versions and parameters used in the reproducibility package

Component	Configuration	Role
Environment	Python 3.11; BERTopic 0.16.4; UMAP 0.5.12; HDBSCAN 0.8.44; NumPy 1.26.4; scikit-learn 1.9.0	Version-pinned analytical environment
Embedding	SPECTER2 base checkpoint (`allenai/specter2_base`); 768 dimensions	Semantic document representation
UMAP for clustering	`n_neighbors` = 15; `min_dist` = 0.1; cosine metric; 5 dimensions; random seed 42	Nonlinear dimensionality reduction before density clustering
HDBSCAN	`min_cluster_size` = 10; `min_samples` = None; Euclidean metric; EOM cluster selection	Density-based cluster discovery with explicit noise
Topic representation	1–2 grams; `min_df` = 2; English stopword set plus documented generic-term list; c-TF-IDF	Interpretable topic keywords after clustering
Visualization	2D UMAP, seed 42	Map coordinates only; not used to fit clusters
Label proposal	Top 25 representative titles; GPT-4o; expert review and renaming	Post-clustering interpretive label layer; membership fixed beforehand

The initial algorithmic solution is evaluated as a density-based partition with explicit noise. HDBSCAN assigns observations outside sufficiently dense semantic regions to -1, and the reference map retains those observations as unassigned. This decision prioritizes specificity of the stable thematic cores while preserving the outlier population for sensitivity analysis. Historical variation in metadata richness may contribute to non-assignment; the noise label is therefore interpreted as a density outcome rather than a data-quality category.

3.3. Documented expert curation

The initial algorithmic topic solution serves as the reference state for subsequent expert curation. Two kinds of curation were applied after clustering: structural operations on topic membership and interpretive operations on topic labels. The reproducibility package stores both the initial algorithmic state and the final curated state, together with a document-level diff, so the intervention and its consequences can be inspected directly.

Structural curation addressed two identifiable editorial situations. One broad initial cluster was split into two analytical lines, while four initial clusters corresponding to segments of the Kipu press/documentary series were consolidated into one documentary topic. The criteria were based on representative titles, semantic content and known editorial-series provenance. The exact number of affected documents and the changes in global structure are reported in the Results because they are outcomes of applying the curation protocol, not method definitions.

Topic labels were generated after membership was defined. GPT-4o proposed labels from the 25 most representative titles per topic, after which expert review retained or renamed them. The final state records whether each label is LLM-derived or manual. The LLM thus assists interpretation only after cluster membership is fixed. This limited role and the retained source evidence follow recent calls for evidence access and verification in LLM-supported scientometric workflows [37].

3.4. Longitudinal structure and internal thematic profile

We characterize the catalogue at two levels: overall thematic composition and line-level trajectories. For each year or analysis period, let n_{kt} be the number of mapped books assigned to topic k, $N_t = \sum_k n_{kt}$ and $p_{kt} = n_{kt}/N_t$. Three overall composition measures are used:

$$HHI_t = \sum_k p_{kt}^2, \tag{1}$$

$${}^1D_t = \exp\left(-\sum_k p_{kt} \ln p_{kt} \right), \tag{2}$$

$${}^2D_t = \frac{1}{HHI_t}, \tag{3}$$

$$d(t,t') = 1 - \frac{\mathbf{p}_t \cdot \mathbf{p}_{t'}}{\|\mathbf{p}_t\| \, \|\mathbf{p}_{t'}\|}. \tag{4}$$

HHI_t measures concentration. ${}^1D_t$ is the Hill diversity of order 1 (the exponential of Shannon entropy), whereas ${}^2D_t = 1/HHI_t$ is Hill diversity of order 2 [38,39]. Reporting both orders prevents the Shannon-effective count from being read as the reciprocal of HHI. The cosine distance $d(t,t')$ measures compositional change between period vectors. All four quantities are interpreted specifically as properties of mapped thematic composition.

The analysis uses two complementary time schemes. The first is the 1978–1986 prelude and the five interpretive periods introduced in Section 3.1 (1987–1998, 1999–2008, 2009–2014, 2015–2021 and 2022–2026), which describe historically meaningful phases of the catalogue and are used to report volumes and mapped books by period. The second is a standardized sequence of adjacent, non-overlapping windows used only to compute compositional distance: the 1978–1986 prelude followed by eight five-year blocks, from 1987–1991 to 2022–2026. Each distance $d(t,t')$ compares one window with the next, so the last value in the series compares 2017–2021 with 2022–2026. In the complete-year sensitivity check, the final window is replaced by 2022–2024.

For line-level analysis, five continuous dimensions summarize internal thematic strength. Table 4 defines them. For topic k, let n_k denote total mapped books, d_k the number of active decades among the five long-run decades, a_k the number of active years and $\ell_k = y_k^{last} - y_k^{first} + 1$ the observed span. Let R_k and B_k denote topic-k book counts in 2015–2026 and 2003–2014, respectively. Volume is relative topic size; persistence records activity across decades; continuity measures the share of active years within the observed span; recent growth combines the minimum recent-volume threshold with the recent-to-previous ratio; and within-family prominence compares a topic's volume with the largest topic in its macro-family. Each dimension is scaled to 0–100 for display but retains its own interpretation.

Table 4. Continuous internal thematic-profile dimensions with their symbols, operationalization and interpretation

Dimension	Symbol	Operationalization	Interpretation
Volume	V	$100\, n_k / \max_j n_j$	Relative catalogue accumulation
Persistence	P	$100\, d_k/5$; d_k = active decades	Long-run temporal presence
Continuity	C	$100\, a_k/\ell_k$; active years / observed span	Regularity within the trajectory
Recent growth	G	$100 \min(R_k/10, 1)$ $\times \min((R_k/\max(B_k, 1))/1.5, 1)$	Screening signal for subsequent renewal qualification
Within-family prominence	F	$100\, n_k / \max_{j \in F(k)} n_j$	Relative volume within the macro-family

The macro-families are obtained by cutting the BERTopic hierarchy at distance threshold 5.0. Sensitivity was checked around this cut: thresholds 4.0, 4.25, 4.5, 4.75, 5.0, 5.25 and 5.5 produce 12, 11, 10, 9, 9, 8 and 7 families, respectively. The nine-family solution is therefore locally stable over the 4.75–5.0 interval. Family membership is determined by the hierarchy, and the family names provide its interpretive labels.

3.5. Two-stage classification of substantive recent renewal

Substantive thematic renewal is operationalized through two sequential conditions: an algorithmic expansion signal followed by trajectory qualification. Let $R_k^{15–26}$ and $R_k^{03–14}$ denote topic-k book counts in 2015–2026 and 2003–2014, respectively, and let y_k^{first} and y_k^{last} denote the first and last active years. The original workflow defined an emergent topic as one whose first active year is 2013 or later. Stage A flags an expansion candidate when

$$\text{Expansion}_k = [R_k^{15–26} \geq 10] \wedge \left[\frac{R_k^{15–26}}{\max(R_k^{03–14}, 1)} \geq 1.5\right] \wedge [y_k^{first} < 2013]. \tag{5}$$

The emergence exclusion affects two of the 68 final topics: *Decolonial pedagogies and insurgent practices* (first active year 2013) and *Yánkuam diaries in Peru and Ecuador* (2018). Both remain part of the thematic map and are excluded only from the expansion-versus-renewal classification because their observed trajectories begin within the recent-growth era.

Stage B was retrospectively formalized from exclusion reasons documented during the original analytical workflow. The resulting rule makes the recorded case decisions executable independently of topic names or IDs. Let I_k^{time} equal 1 when no documented chronological anomaly compromises the series; let $q_k^{(3)}$ be the share of all topic-k books concentrated in its three highest-output years; let $\ell_k = y_k^{last} - y_k^{first} + 1$ be the observed span; and let n_k be total mapped books in the topic. Episodicity is defined as

$$\text{Episodic}_k = [q_k^{(3)} \geq 0.60] \wedge [\ell_k \geq 10] \wedge [n_k \geq 15]. \tag{6}$$

A reference substantive-renewal classification is then

$$\text{Renewal}_k = \text{Expansion}_k \wedge I_k^{time} \wedge \neg\text{Episodic}_k \wedge [y_k^{last} \geq 2022]. \quad (7)$$

The recency gate uses 2022 because 2022–2026 is the independently defined recent recomposition period. The chronological-integrity condition is evaluated upstream; after correction for the Kipu registration-date anomaly, the frozen Stage-A set contains eight candidates. Because Stage B was formalized ex post, the four-topic outcome is treated as the reference-case classification and its threshold dependence is evaluated explicitly. Sensitivity is evaluated over 36 combinations of the growth-ratio threshold (1.25, 1.50, 1.75), episodicity threshold (0.50, 0.55, 0.60, 0.65) and final-year gate (2021, 2022, 2023).

3.6. OpenAlex indexed scientific activity

Nine editorial lines were selected for the external proof-of-concept: six lines with strong historical convergence in the internal catalogue and three additional lines identified through substantive recent renewal. Amazonian literature and culture is itself a renewed line but was already present in the historical block, so it does not increase the nine-line set. The nine lines were mapped to 46 line-topic correspondences involving 40 unique OpenAlex topics. Each correspondence was classified by its role in representing the editorial line: principal (weight 1.00), complementary (0.65), specific subline (0.55) or contextual/partial (0.35).

For each OpenAlex topic o and country c, five signals were computed: production in 2015–2025, recent production in 2021–2025, world share, international collaboration and growth trend. Signals were normalized within the ten pre-specified countries for each topic. The production component uses logarithmic transformation before normalization. The topic-country score has an effective maximum of 90 points:

$$Q_{oc} = 30\tilde{N}_{oc}^{15\text{–}25} + 25\tilde{N}_{oc}^{21\text{–}25} + 15\tilde{W}_{oc} + 10\tilde{C}_{oc} + 10\tilde{T}_{oc}. \quad (8)$$

A line-country score is the role-weighted mean of the mapped topic-country scores, rescaled once from 90 to 100,

$$A_{\ell c} = \frac{100}{90} \frac{\sum_{o \in O_\ell} w_{\ell o} Q_{oc}}{\sum_{o \in O_\ell} w_{\ell o}}. \quad (9)$$

The ten-country comparison comprises Ecuador, Mexico, Spain, Colombia, Peru, Argentina, Bolivia, Brazil, Chile and the United States. This set was pre-specified before the present comparative analysis and is retained as a bounded comparison set. Because OpenAlex coverage and metadata quality vary by language, field and indexing infrastructure, country scores are conditional on database representation; evidence from Latin America suggests broader Spanish-language and SSH coverage than Scopus in some settings, while independent audits also identify language-metadata errors [31,32]. The equal-country comparison serves as the scientific reference scenario. A second, alternative strategic scenario weights Ecuador–Mexico–Spain at 60% and the other seven countries at 40% when a line-level territorial summary is needed. Robustness checks perturb the component and mapping-role weights and compare the strategic scenario with the equal-country territorial pattern.

3.7. Research Horizon screening and documentary review

Clarivate Research Horizon Navigator serves as a fixed exported source of candidate emerging-topic clusters; research-front relevance is subsequently qualified through corpus-wide documentary review. The exported universe available to the study contained 11,325

topic IDs corresponding to 11,232 unique names. Candidate generation combined the nine internal editorial profiles, OpenAlex bridge topics, expanded textual retrieval and manual review. This structured expert-assisted screening produced 73 line-cluster associations in three screening tiers: 30 Priority 1, 16 Priority 2 and 27 held in a screening reserve (a tier distinct from the "reserve" outcome of the documentary review described below). The screening files also preserve false-positive examples, which are important because lexical proximity alone sometimes retrieves clusters that are regionally or substantively unrelated to the editorial line.

A preliminary 100-point affinity heuristic supported prioritization of candidate datasets: thematic correspondence (45 points), bridge with OpenAlex (25), fit with the editorial line (20) and regional/cultural signal (10). The heuristic did not determine the final frontier category. All 30 Priority-1 datasets were obtained, comprising 1,084 paper records. The recorded dataset sizes match the consolidated record-level corpus exactly. Titles, abstracts, keywords, subject composition, affiliations and representative papers were then reviewed to classify each candidate as principal, complementary, specific subline, reserve or excluded.

For each editorial line ℓ, the primary research-front output is the retained-candidate fraction

$$H_\ell = \frac{N_\ell^{principal} + N_\ell^{complementary} + N_\ell^{specific}}{N_\ell^{reviewed}}, \tag{10}$$

reported together with the role composition of retained candidates. The numerator includes principal, complementary and specific-subline candidates, while reserve and excluded cases remain outside the retained set. H_ℓ is the retention fraction within the Priority-1 set screened for each line, whose denominator ranges from two to six. It answers the question "what share of the screened candidate set survived documentary review?" while showing the line-specific screening intensity in the denominator.

The screening records preserve the full path from the 11,325-topic export through the candidate associations to the documentary review of the 30 Priority-1 corpora, including the reserve and excluded cases, so that every classification can be traced to its documentary basis.

3.8. Reproducibility, robustness and evaluation strategy

Deterministic and interpretive components have different reproducibility conditions. The topic-model package contains the 3,963 model records, frozen inputs, 768-dimensional embeddings, original five- and two-dimensional UMAP arrays, initial reference clusters, final curated state, parameter files, a version-pinned environment, hashes, notebooks and verification scripts. The anchored five-dimensional UMAP representation allows the clustering state to be checked against stored assignments. Re-projecting embeddings uses a fixed seed but can still vary across software and numerical environments, so the archived arrays define the computational reference. LLM-generated labels and expert decisions are fixed interpretive records rather than deterministic model outputs. This distinction follows the principle that multi-step bibliometric workflows should expose computational operations and input dependencies [37,40]. All reported quantities derive from executed scripts over the frozen data and analytical states.

The robustness design targets the main analytical decisions:

- First, initial algorithmic and curated thematic states are compared directly.
- Second, the macro-family cut is varied around the selected threshold.
- Third, assignment rates and simple preserved-text characteristics are compared for assigned and unassigned records.

- Fourth, outlier sensitivity is examined in two ways using the frozen SPECTER2 embeddings. An intentionally aggressive stress test assigns every HDBSCAN noise record to its nearest final-topic cosine centroid. A controlled high-confidence scenario reassigns only noise records that exceed the 25th-percentile thresholds of both own-topic centroid similarity and assignment margin observed among mapped records. The reference map remains the primary specification, while these scenarios quantify the dependence of downstream diversity, compositional distance and renewal membership on the outlier policy.
- Fifth, the two-stage renewal rule is perturbed over 36 threshold combinations.
- Sixth, the 2017–2021 to 2022–2026 compositional change is recomputed using the complete 2022–2024 subwindow.
- Seventh, HDBSCAN minimum-cluster-size sensitivity is explored on the archived five-dimensional representation using a cross-implementation perturbation; the reference-like setting is first checked against the frozen initial algorithmic partition with ARI and NMI. The frozen execution environment does not support an exact historical UMAP `n_neighbors` rerun, so neighborhood coherence is inspected at 10, 15, 20 and 30 nearest neighbors in the original 768-dimensional embedding space. This provides a semantic-neighborhood-scale diagnostic that is analytically distinct from a UMAP-parameter sensitivity test.
- Finally, the OpenAlex geographic conclusions are compared under equal-country and 60/40 territorial scenarios and under 5,000 Monte Carlo perturbations of component and mapping-role weights.

Detailed results appear in Appendix A, and the Data Availability Statement summarizes the reproducibility package.

4. Results: longitudinal proof-of-concept

The proof-of-concept follows the analytical sequence: coverage and stability of the internal thematic map, long-run change, profiles of the nine focal lines, indexed scientific activity and research-front retention. Section 4.7 and the Discussion then interpret the sources jointly.

4.1. Corpus coverage and the effect of documented curation

The catalogue contained 4,142 validated book records after consolidation (Table 5). The text-eligibility rule retained 3,963 records for semantic modelling; 179 validated books did not meet either the title-length or abstract-length condition. HDBSCAN assigned 2,383 of the semantically eligible books to stable density regions and left 1,580 as noise. The resulting assignment rate is 60.1% among semantically eligible books and 57.5% relative to the full validated catalogue. These percentages define the evidential scope of all topic-level results below: the thematic map describes the stable semantic cores identified by the model, not every validated book in the catalogue.

The non-assigned share is substantial, but it is not concentrated in a single historical decade. Among semantically eligible records, assignment rates were 65.2% in 1978–1989, 64.0% in the 1990s, 59.7% in the 2000s, 58.2% in the 2010s and 59.2% in 2020–2026. The decline from the earliest periods is moderate rather than discontinuous. This pattern is consistent with the interpretation of HDBSCAN noise as the complement of stable density regions.

The initial algorithmic partition contained 70 clusters before expert curation. The documented structural curation produced 68 final analytical topics while preserving the same 2,383 assigned books. A document-level comparison identifies 389 records whose topic identifier changed, equivalent to 9.82% of the 3,963 semantically eligible records.

Table 5. Corpus funnel from the validated catalogue to the stable topic assignments

Stage	Records	Share	Interpretation
Validated catalogue	4,142	100.0% of validated books	Documentary universe used for descriptive counts
Semantically eligible	3,963	95.7% of validated books	Title $\geq$ 20 characters OR abstract $\geq$ 200 characters
Assigned to stable topics	2,383	60.1% of eligible books	Basis of topic, family, continuity and composition analyses
HDBSCAN noise	1,580	39.9% of eligible books	Retained as unassigned; no nearest-topic forcing
Outside semantic eligibility	179	4.3% of validated books	Remain part of the validated catalogue but not of the semantic model

The intervention therefore affected a limited subset rather than replacing the algorithmic structure wholesale. It changed the global distribution modestly (Table 6): concentration decreased from $HHI = 0.0316$ to 0.0259, while Hill diversity of order 1 increased from 49.83 to 51.10. In the curated map, Hill diversity of order 2 is 38.60. These changes follow from splitting a broad mixed cluster and consolidating the four Kipu segments into a documentary-series topic.

Table 6. Comparison of the initial algorithmic partition and the expert-curated thematic state

Measure	Initial algorithmic state	Final curated state	Reading
Topics/clusters	70	68	One broad cluster split; four Kipu clusters merged into one documentary topic
Assigned books	2,383	2,383	Curation changes organization, not the assigned-book denominator
Documents with changed topic identifier	–	389	9.82% of semantically eligible records
Herfindahl concentration	0.0316	0.0259	Final map is slightly less concentrated
Hill diversity ^1D	49.83	51.10	Shannon-effective topic diversity increases slightly

This comparison answers RQ2 by quantifying the role of curation rather than treating expert intervention as an undocumented correction.

The outlier-reassignment stress tests quantify how the 39.9% noise share affects downstream results. Forcing all 1,580 noise records to the nearest final-topic centroid changes HHI from 0.0259 to 0.0242, Hill diversity ^1D from 51.10 to 53.01 and ^2D from 38.60 to 41.33. The global diversity structure is therefore comparatively stable. Temporal conclusions are more sensitive: the 2017–2021 to 2022–2026 cosine distance falls from 0.363 to 0.258 under complete forcing. A controlled high-confidence scenario reassigns only 166 noise records and yields $^1D = 51.34$, $^2D = 39.08$ and a recent distance of 0.334. Thus, the evidence supports a recent compositional shift across scenarios, but its magnitude depends on whether diffuse records are forced into stable topics.

The preserved fields do not support a complete diagnosis of why records became noise, but they rule out one simple explanation. Unassigned records do not have shorter titles: the median original-title length is 66 characters among noise records versus 60 among

assigned records; the corresponding model-text medians are 64 and 59. Exact duplicate-title shares are also similar (12.9% vs. 12.5%).

4.2. The final thematic landscape: distributed concentration and long persistence

The final map contains 68 topics organized into nine hierarchical macro-families. Figure 3 displays the mapped books in the two-dimensional UMAP visualization, with topic centroids superimposed and the nine focal lines labelled; the frame is cropped to the dense region, which leaves the 26 books and the centroid of the single-topic family I outside it. The two-dimensional coordinates are used only for visualization; topic membership derives from the five-dimensional clustering representation described in Section 3.2.

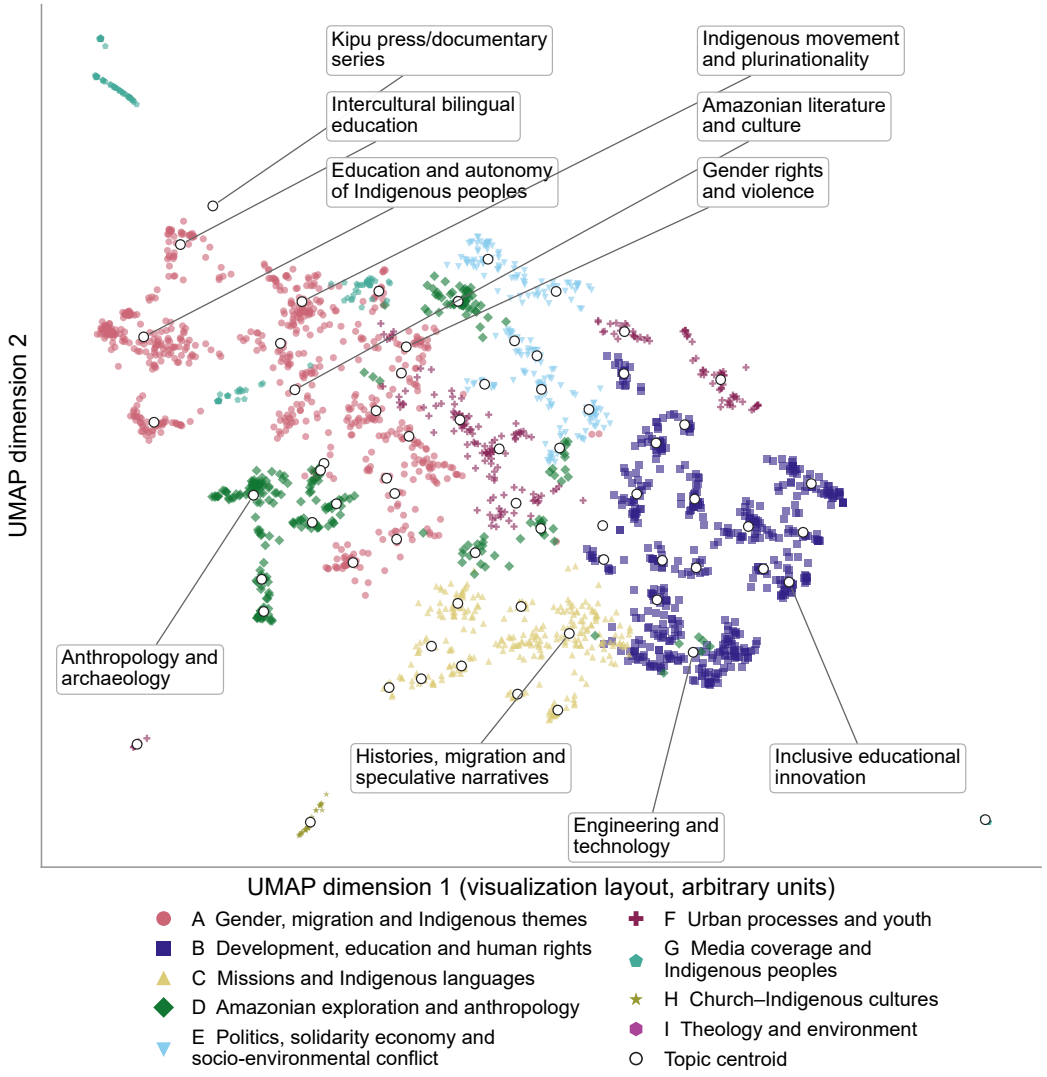


Figure 3. Thematic map of the curated catalogue in the two-dimensional UMAP layout. Colour and marker shape identify the nine macro-families; open circles mark topic centroids.

The topic-size distribution is broad rather than dominated by a few clusters (Table 7). The final *HHI* of 0.0259 corresponds to 51.10 effective topics, meaning that the observed concentration is comparable to a hypothetical distribution with about 51 equally common topics. The largest topic, *Education and autonomy of Indigenous peoples*, contains 150 books (6.29% of mapped books), followed by *Engineering and technology* with 147 (6.17%) and *Histories, migration and speculative narratives* with 137 (5.75%). The five largest topics together account for 26.5% of the mapped corpus, leaving nearly three quarters distributed across the remaining 63 topics.

Long persistence is nevertheless widespread. Forty-six topics are active in at least four decades and together contain 1,952 books, or 81.9% of mapped records. Eighteen topics appear in all five long-run decades used for the persistence calculation and account for 1,157 books (48.6%). Persistence should not be read as uniform annual continuity: for example, *Education and autonomy of Indigenous peoples* is active in 35 of the 37 years between its first and last mapped records, whereas *Amazonian literature and culture* is active in 27 of 38 years. These distinctions motivate retaining persistence and continuity as separate profile dimensions.

Table 7. Ten largest final topics with their size, observed span and temporal persistence

Topic	Books	Share	Observed span	Active years	Active decades
Education and autonomy of Indigenous peoples	150	6.29%	1987–2023	35	5
Engineering and technology	147	6.17%	1987–2024	32	5
Histories, migration and speculative narratives	137	5.75%	1978–2025	38	5
Kipu press/documentary series	105	4.41%	1983–2024	40	5
Indigenous movement and plurinationality	92	3.86%	1989–2026	33	5
Anthropology and archaeology	76	3.19%	1988–2026	28	5
Amazonian literature and culture	65	2.73%	1987–2024	27	5
Interculturality and cultural educational management	61	2.56%	1989–2025	25	5
Political culture and democracy	60	2.52%	1989–2025	26	5
Community development and university extension	56	2.35%	2007–2023	14	3

The nine macro-families are an interpretive higher-order organization rather than a substitute for the 68-topic map. Their sizes range from 681 mapped books in the largest family to 26 in the smallest. Varying the hierarchy cut confirms that the nine-family solution is locally stable but not uniquely determined: the number of families changes from 12 at threshold 4.0 to seven at 5.5, remaining at nine for thresholds 4.75 and 5.0. We therefore use the family layer to provide relational context and to calculate within-family prominence, not to claim a uniquely true taxonomy of the catalogue.

4.3. Longitudinal evolution: historical accumulation and recent recomposition

The catalogue grows unevenly over the observation period. Figure 4 shows the number of validated books per catalogue year, split into books assigned to stable topics and the remaining validated books (HDBSCAN noise or records outside text eligibility), with the interpretive periods shaded, the missing 1979 record indicated and the incomplete 2025–2026 tail hatched; Table 8 reports the totals by period. The 1978–1986 prelude contains only 17 validated books and 13 mapped books. Volume subsequently expands: 1987–1998 contains 870 validated and 513 mapped books; 1999–2008 contains 1,001 validated and 551 mapped books; and 2009–2014 reaches the maximum six-year volume with 1,073 validated and 612 mapped books. The annual count of validated books peaks at 242 in 2013. The following windows contain 844 validated and 488 mapped books in 2015–2021 and 337 validated and 206 mapped books in 2022–2026. The last window includes partial 2025–2026 catalogue coverage and is therefore interpreted primarily as a composition signal rather than a finalized volume estimate.

Compositional change varies independently of total output (Figure 5; the windows contain 13, 172, 224, 323, 251, 367, 500, 327 and 206 mapped books). The transition from the small prelude to 1987–1991 has the largest raw distance (0.679), as expected from the very small initial base. Excluding that discontinuity, the 2017–2021 to 2022–2026 transition

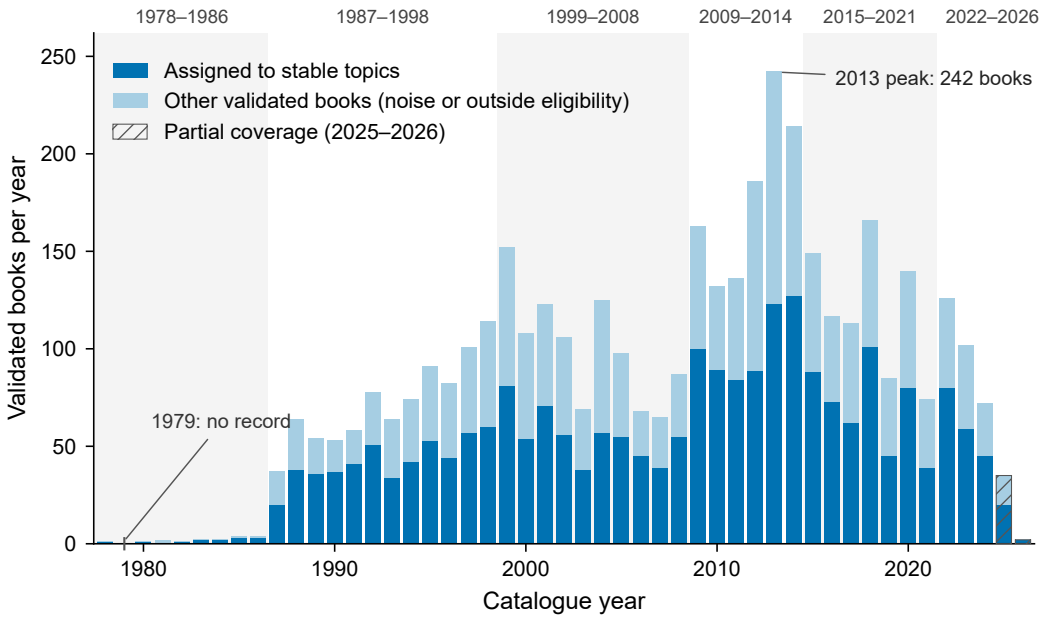


Figure 4. Validated books per catalogue year, 1978–2026, split into books assigned to stable topics and the remaining validated books. Shaded bands mark the interpretive periods.

Table 8. Validated and mapped books by interpretive period with their analytical reading

Period	Validated books	Mapped books	Mapped / validated	Analytical reading
1978–1986	17	13	76.5%	Documentary prelude; too small for fine trend inference
1987–1998	870	513	59.0%	Foundation and consolidation of long-run lines
1999–2008	1,001	551	55.0%	Diversified plateau
2009–2014	1,073	612	57.0%	Maximum catalogue expansion
2015–2021	844	488	57.8%	Contraction and transition
2022–2026	337	206	61.1%	Recent recomposition; 2025–2026 coverage is partial

has the largest adjacent-window distance, 0.363. Earlier post-foundation distances range from 0.168 to 0.333. Restricting the recent window to the complete 2022–2024 years yields a slightly larger distance of 0.371, so incomplete 2025–2026 coverage does not account for the recent shift. Under the controlled high-confidence outlier reassignment, the corresponding distances are 0.334 and 0.341. The recent catalogue therefore differs from the 2010s portfolio, although the magnitude of recomposition depends on how diffuse records are treated. The direction of change is selective rather than wholesale because many large historical topics remain active while a limited group of smaller lines increases its recent share.

This pattern provides the first answer to RQ1. Long-run thematic structure is characterized simultaneously by high persistence, distributed topic diversity and episodic changes in composition, which are captured by separate longitudinal statistics.

4.4. Internal thematic profiles and the distinction between growth and substantive renewal

Figure 6 compares the five continuous internal dimensions across the nine focal editorial lines, and Table 9 reports the values. Historically accumulated lines have different internal signatures. *Education and Indigenous autonomy* combines maximum relative volume

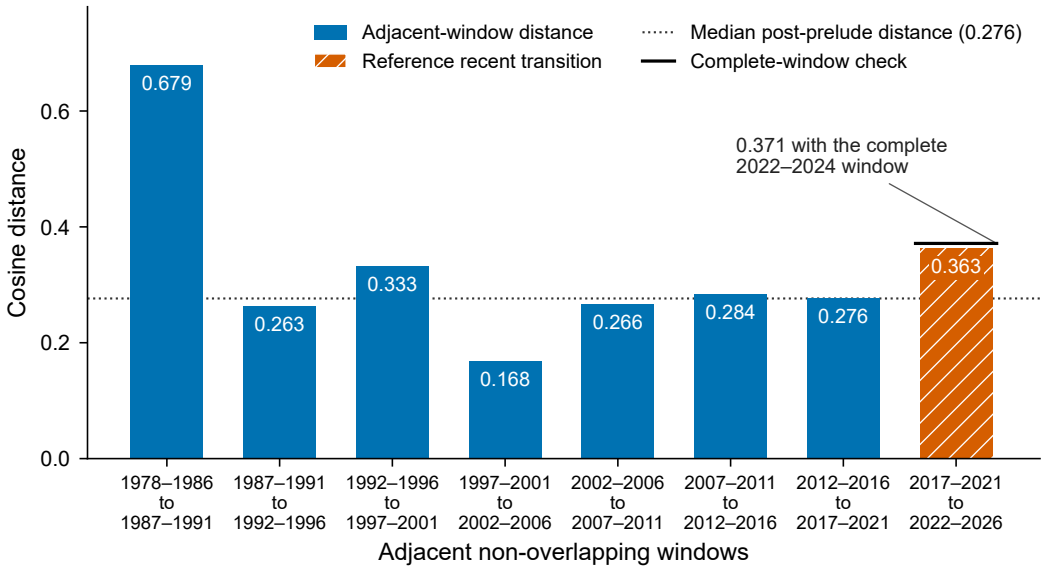


Figure 5. Cosine distance between the topic compositions of adjacent non-overlapping windows. The hatched bar is the 2017–2021 to 2022–2026 transition; the black tick is the same transition with the complete 2022–2024 window.

and persistence with 94.6 on continuity and within-family prominence of 100. *Engineering and technology* is nearly as large (98.1 on relative volume) and persists across all five decades, but its recent-growth signal is only 32.5 because its 2015–2026 output is lower relative to the 2003–2014 base. *Histories, migration and speculative narratives* also combines high volume and long persistence, while its continuity is lower than that of the education line.

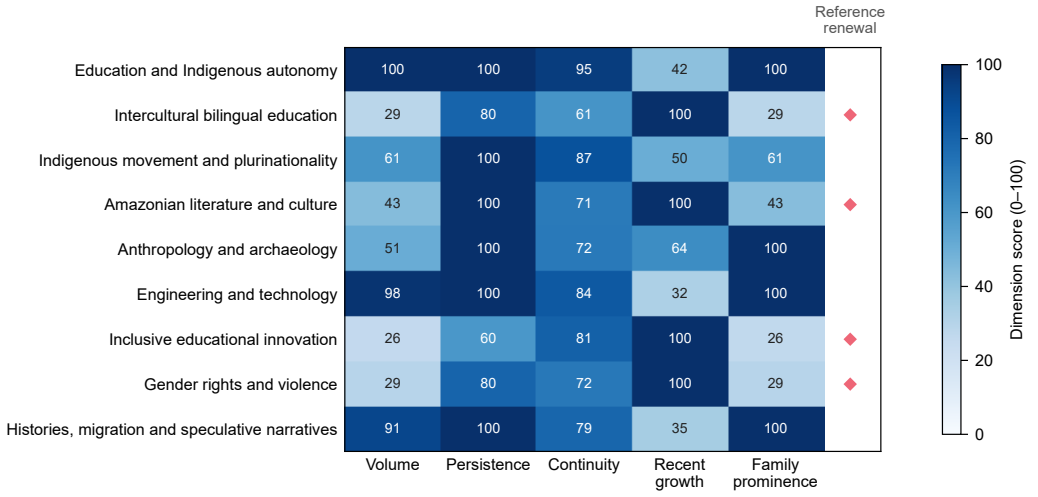


Figure 6. Internal thematic profiles of the nine focal lines (0–100 display scale). Diamonds mark the reference renewals.

The three lines added to the external analysis because of renewal have a contrasting profile. *Inclusive educational innovation* has a relative volume of 26.1 and persistence of 60, but continuity of 81.2 and the maximum recent-growth signal. *Gender rights and violence* and *Intercultural bilingual education* also have modest accumulated volume but maximum recent-growth scores. Amazonian literature and culture is the fourth renewed line; it was already included among the historically relevant lines and therefore does not add a tenth focal line. The profiles show that a relatively small stock can coexist with a strong current trajectory, while a very large historical stock can coexist with weaker recent growth.

Table 9. Five-dimensional internal profile and renewal status of the nine focal editorial lines

Line	Books	*V*	*P*	*C*	*G*	*F*	Renewal
Education and Indigenous autonomy	150	100.0	100	94.6	41.7	100.0	No
Intercultural bilingual education	43	28.6	80	61.1	100.0	28.7	Yes
Indigenous movement and plurinationality	92	61.4	100	86.8	50.0	61.3	No
Amazonian literature and culture	65	43.4	100	71.1	100.0	43.3	Yes
Anthropology and archaeology	76	50.7	100	71.8	64.3	100.0	No
Engineering and technology	147	98.1	100	84.2	32.5	100.0	No
Inclusive educational innovation	39	26.1	60	81.2	100.0	26.5	Yes
Gender rights and violence	44	29.4	80	71.9	100.0	29.3	Yes
Histories, migration and speculative narratives	137	91.4	100	79.2	34.9	100.0	No

Two final topics are classified as emergent under the first-active-year cutoff: *Decolonial pedagogies and insurgent practices* (2013) and *Yánkuam diaries in Peru and Ecuador* (2018). They remain in the thematic map and are handled separately from the expansion screen. Stage A of the renewal procedure identifies eight expansion candidates among the non-emergent topics, and Stage B reduces these to four through topic-independent decision rules (Table 10). Figure 7 makes the logic visible. Three candidates—urban social inclusion, digital media and social networks and citizen/multidimensional communication—have large recent-to-previous ratios but concentrate at least 60% of their output in only three years, satisfying the pre-existing episodicity rule. Socio-environmental mining conflicts is not episodic, but its last mapped book is from 2020 and it therefore fails the 2022 recency gate. The four surviving lines are Amazonian literature and culture (38 vs. 13 books, ratio 2.92), inclusive educational innovation (27 vs. 12, ratio 2.25), intercultural bilingual education (19 vs. 12, ratio 1.58) and gender rights and violence (24 vs. 16, ratio 1.50).

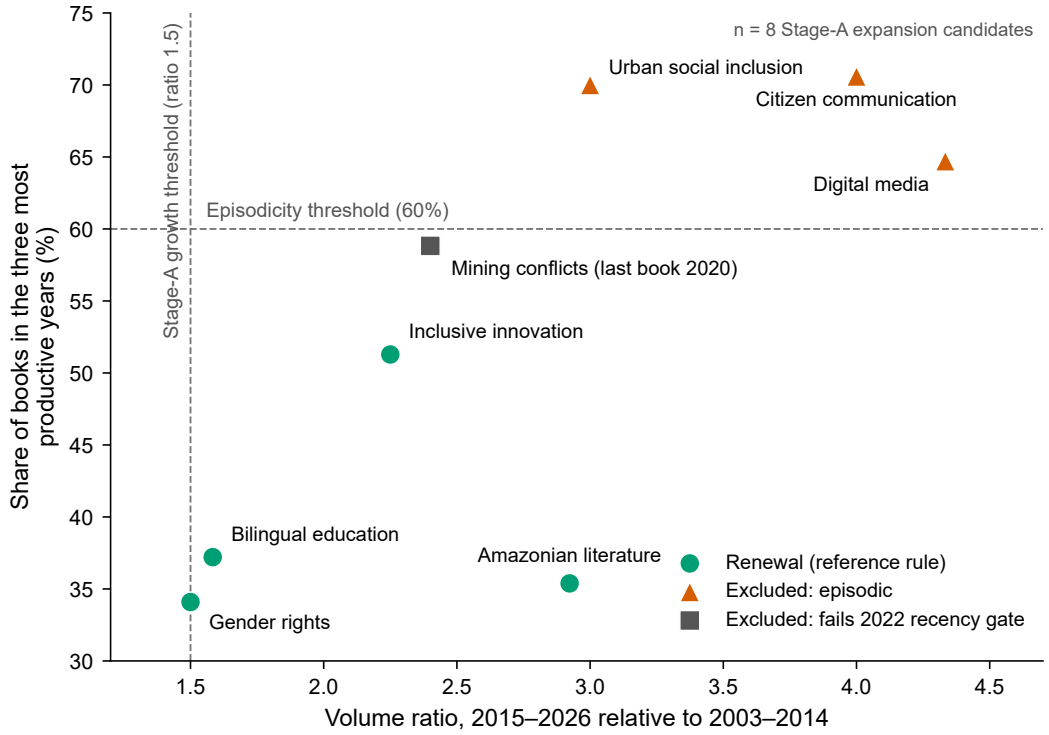


Figure 7. Two-stage renewal classification of the eight expansion candidates. Dashed lines mark the growth-ratio and episodicity thresholds; colour and marker shape give the recorded decision.

Table 10. Audit trail of the two-stage renewal classification for the eight expansion candidates

Candidate	2015–2026	2003–2014	Ratio	Top-3-year share	Stage-B outcome
Amazonian literature and culture	38	13	2.92	35.4%	Renewal; active to 2024
Gender rights and violence	24	16	1.50	34.1%	Renewal; active to 2025
Intercultural bilingual education	19	12	1.58	37.2%	Renewal; active to 2025
Inclusive educational innovation	27	12	2.25	51.3%	Renewal; active to 2024
Urban social inclusion	15	5	3.00	70.0%	Excluded as episodic
Digital media and social networks	13	3	4.33	64.7%	Excluded as episodic
Citizen and multidimensional communication	12	3	4.00	70.6%	Excluded as episodic
Socio-environmental mining conflicts	12	5	2.40	58.8%	Excluded; last active year 2020

The comparison resolves a potential ambiguity in the term "expansion." A fourfold increase is not necessarily substantive renewal when production is highly concentrated in a few years. Conversely, a ratio of 1.50 can qualify under the reference rule when the trajectory is non-episodic and remains active in the recent recomposition period. The four reference renewals are preserved under the controlled high-confidence outlier reassignment, but Stage-B threshold perturbation shows that exact membership is not invariant. Across 36 combinations of the growth-ratio threshold (1.25–1.75), episodicity threshold (50–65%) and final-year gate (2021–2023), the number of renewals ranges from one to seven. Amazonian literature and culture is retained in all scenarios, inclusive educational innovation in 75.0% and gender rights and intercultural bilingual education in 66.7% each. RQ1 is therefore answered by an executable reference classification together with an explicit account of its threshold dependence.

4.5. Indexed scientific activity across focal lines and countries

OpenAlex produces a different view because the unit of comparison is scientific activity around mapped topics rather than books in the ABYA-YALA catalogue. Figure 8 reports the 90 line-country scores. Averaged over the nine lines, the United States has the highest comparative score among the ten pre-specified countries (71.47), followed by Spain (53.74), Brazil (52.36), Mexico (42.94), Argentina (36.99) and Chile (36.41). Ecuador's mean is 23.51 and Bolivia's 9.06. These values describe the relative configuration of the selected OpenAlex topics within this bounded ten-country comparison.

The geography varies by editorial line (Table 11). The United States has the highest score for seven of the nine lines, including engineering and technology (93.62), histories and migration (86.63), gender rights and violence (78.80), intercultural bilingual education (74.32) and inclusive educational innovation (74.43). Two lines differ. Argentina leads Indigenous movement and plurinationality with 67.88, ahead of Spain (61.72) and Brazil (61.08). Brazil leads Amazonian literature and culture with 62.14, narrowly ahead of the United States (60.97) and Spain (60.84). The result therefore supports a topic-by-territory reading rather than a single geographical hierarchy.

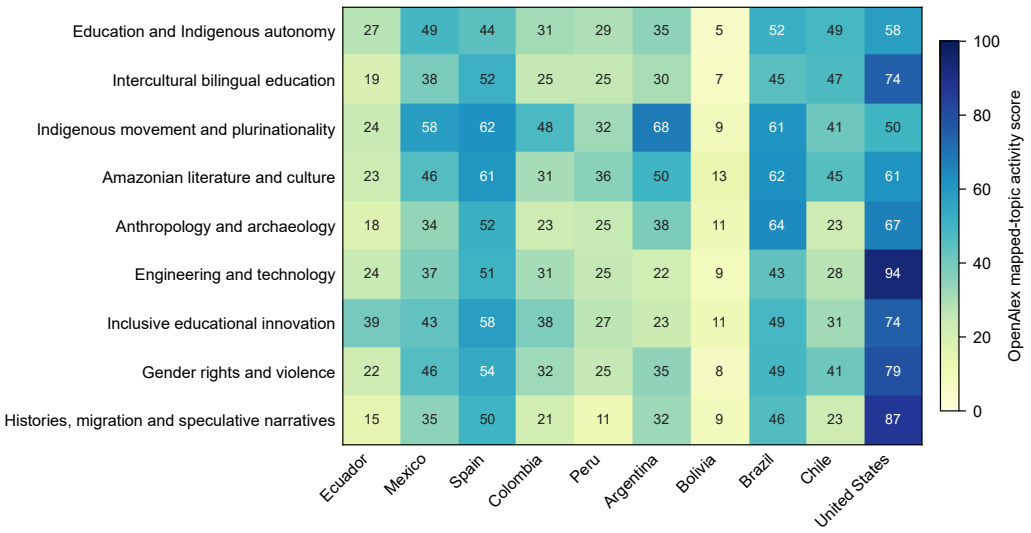


Figure 8. OpenAlex indexed-activity scores for the nine focal lines across the ten pre-specified countries.

Table 11. Highest OpenAlex line-country mapped-topic activity scores within the ten-country comparison

Editorial line	Highest country	Score	Second country	Score
Education and Indigenous autonomy	United States	58.09	Brazil	52.12
Intercultural bilingual education	United States	74.32	Spain	51.51
Indigenous movement and plurinationality	Argentina	67.88	Spain	61.72
Amazonian literature and culture	Brazil	62.14	United States	60.97
Anthropology and archaeology	United States	66.52	Brazil	63.72
Engineering and technology	United States	93.62	Spain	50.99
Inclusive educational innovation	United States	74.43	Spain	58.09
Gender rights and violence	United States	78.80	Spain	53.99
Histories, migration and speculative narratives	United States	86.63	Spain	50.46

The strategic 60/40 territorial scenario does not materially reorder the nine lines relative to a simple equal-country mean. The Spearman rank correlation between the two line-level summaries is $\rho = 0.983$, and the largest absolute score change is 3.21 points, for inclusive educational innovation. A separate 5,000-run perturbation of component and mapping-role weights leaves the United States first in the overall ten-country portfolio in every run; Spain and Brazil occupy second or third place in all simulations. These checks do not eliminate dependence on the chosen country set, but they show that the main comparative pattern is not a consequence of a single set of numerical weights.

These findings answer RQ3 within the scope of the pre-specified comparison. Indexed scientific activity around the mapped editorial themes varies substantially by topic and country, and the strongest country-level score depends on the theme. The ten-country set is a bounded, pre-specified comparison rather than a global sample, and OpenAlex coverage is itself heterogeneous. Evidence from Argentina indicates that OpenAlex can recover substantially broader Spanish-language and SSH production than Scopus [31], while a separate audit shows that OpenAlex language metadata can overestimate English and underestimate other languages [32]. The layer therefore provides bounded context on indexed scientific activity around the mapped topics.

4.6. Research-front retention after corpus-wide documentary review

The Research Horizon layer further separates preliminary metadata affinity from document-supported retention. Structured screening produced 30 Priority-1 candidate clusters whose corpora contained 1,084 paper records. Corpus-wide documentary review of titles, abstracts, keywords, subject fields, affiliations and representative papers retained 23 clusters as principal, complementary or specific sublines; four were placed in reserve and three were excluded. Across the 23 retained clusters, seven are principal, six complementary and ten specific sublines.

Table 12 reports, for each of the nine focal lines, the number of Priority-1 candidates reviewed, their classification as principal, complementary, specific subline, reserve or excluded, and the resulting retained fraction.

Table 12. Research Horizon documentary-review outcomes and retained fractions for the nine focal lines

Line	Cand.	Principal	Compl.	Specific	Reserve	Excl.	Retained share
Education and Indigenous autonomy	2	1	1	0	0	0	2/2 (1.00)
Intercultural bilingual education	3	2	1	0	0	0	3/3 (1.00)
Indigenous movement and plurinationality	3	0	0	0	2	1	0/3 (0.00)
Amazonian literature and culture	3	0	1	0	1	1	1/3 (0.33)
Anthropology and archaeology	3	1	0	1	1	0	2/3 (0.67)
Engineering and technology	6	0	0	6	0	0	6/6 (1.00)
Inclusive educational innovation	3	0	1	2	0	0	3/3 (1.00)
Gender rights and violence	4	2	1	1	0	0	4/4 (1.00)
Histories, migration and speculative narratives	3	1	1	0	0	1	2/3 (0.67)

Retention is complete for four focal lines: intercultural bilingual education retains 3/3 candidates, engineering and technology 6/6, inclusive educational innovation 3/3 and gender rights and violence 4/4. Education and Indigenous autonomy also retains both screened candidates (2/2). Anthropology and archaeology and histories/migration each retain 2/3. Amazonian literature and culture retains 1/3. Indigenous movement and plurinationality retains none of its three Priority-1 candidates.

Two cases show why corpus-wide documentary review matters. Indigenous movement and plurinationality had an average preliminary affinity of 91.7/100 across its three candidates, yet two were placed in reserve and one was excluded. Amazonian literature and culture had an even higher preliminary average (94.7) but only one complementary cluster was retained; one candidate remained in reserve and one was excluded. High preliminary affinity therefore does not guarantee that a screened candidate survives documentary review.

These results answer RQ4 for the screened candidate set. The retained fraction keeps candidate-set size visible instead of converting the outcome into an ordinal category whose attainable ceiling would depend on the number of candidates screened. H_ℓ is thus a retention share within the screened set, with a line-specific denominator, not a calibrated probability of proximity to emerging research fronts in general. Because candidate generation was a targeted, expert-assisted selection within the 11,325-topic universe, a low retained fraction indicates that the screened candidates did not survive documentary review for that line, not that no related front exists.

4.7. Cross-layer configurations and answers to the research questions

Figure 9 compares the three sources. The horizontal axis gives the equal-country mean of OpenAlex mapped-topic activity, and the vertical axis gives the Research Horizon retained-candidate fraction. Both use their native descriptive scales; marker shape identifies the four reference renewals.

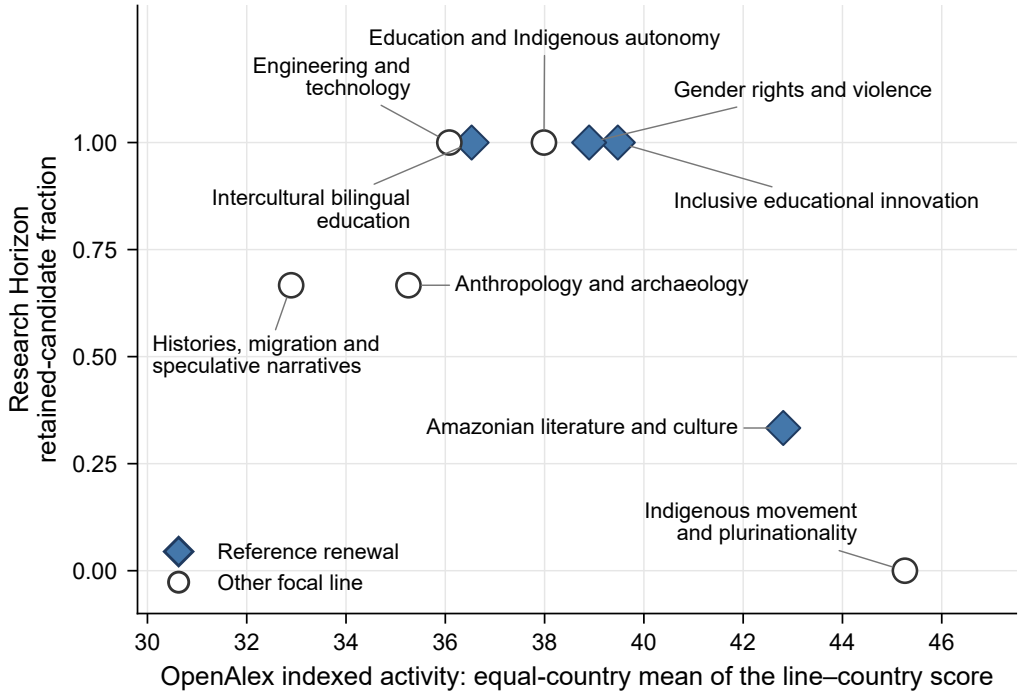


Figure 9. Cross-layer configuration of the nine focal lines: OpenAlex equal-country mean versus Research Horizon retained fraction. Diamonds mark the reference renewals; overlapping markers are offset horizontally.

Three recurring configurations are visible; Table 13 then summarizes the answers to the four research questions.

- First, some lines with high historical volume also show relatively high indexed scientific activity, but their screening outcomes differ. Engineering and technology combines very high internal volume and persistence with the highest United States OpenAlex score and retains all six screened Research Horizon sublines. Anthropology and archaeology has lower internal volume but full persistence, within-family prominence and two of three candidates retained. Education and Indigenous autonomy has the largest internal stock and retains both screened candidates.
- Second, recent renewal can coexist with a smaller historical stock. Intercultural bilingual education, inclusive educational innovation and gender rights and violence have internal volume scores below 30, satisfy the reference renewal rule and retain all screened Research Horizon candidates. Their profiles reflect current trajectories and full retention among screened candidates rather than high historical volume.
- Third, a strong internal profile can diverge from research-front evidence. Indigenous movement and plurinationality has 92 mapped books, activity in all five decades and 86.8 continuity, yet none of its three high-scoring preliminary Research Horizon candidates is retained. Amazonian literature and culture satisfies the reference renewal rule and has a strong Brazil-centered OpenAlex profile, but retains only one of three screened candidates. These cases show that internal thematic strength, indexed scientific activity and research-front retention are not interchangeable.

Table 13. Empirical answers to the four research questions and the main evidence supporting them

RQ	Empirical answer	Main evidence
RQ1	The mapped stable semantic core is diverse, persistent and recently recomposed, while the exact renewal set is partly threshold-sensitive.	68-topic map; $HHI = 0.0259$; ${}^1D = 51.10$; ${}^2D = 38.60$; 46 topics active in at least four decades; reference 8-to-4 renewal classification with threshold sensitivity.
RQ2	Expert curation changes a documented subset of the initial algorithmic structure, and its effects can be measured explicitly.	70 initial algorithmic clusters to 68 analytical topics; 389 records with changed topic ID; HHI 0.0316 to 0.0259; effective topics 49.83 to 51.10.
RQ3	Within the ten-country comparison, indexed scientific activity around mapped topics is strongly topic- and territory-specific and remains stable under the tested weighting alternatives.	90 line-country OpenAlex scores; different leading countries across lines; equal-country vs. 60/40 rank correlation $\rho = 0.983$.
RQ4	Within the screened candidate set, retention after documentary review varies sharply across lines and does not follow historical accumulation or preliminary metadata affinity in a simple way.	23 of 30 Priority-1 clusters retained; retained fractions range from 0/3 to 6/6; movement/plurinationality has high preliminary affinity but no retained research-front match.

5. Discussion

The proof-of-concept supports analyzing a scholarly publisher as a longitudinal thematic system. Its main empirical result is configurational: historical volume, recent renewal, indexed scientific activity and research-front retention align for some editorial lines and diverge for others. These configurations cannot be inferred from catalogue volume alone. The discussion considers this longitudinal unit of analysis, documented expert interpretation and the scope of the two external sources.

5.1. A publisher catalogue as a longitudinal knowledge structure

Bibliometric research has established the distinctive role of monographs and edited volumes in the social sciences and humanities. It has also developed assessments based on publisher prestige, citations, library holdings and publisher-level indicators [1–7,19,22,41]. These approaches remain central to studying impact, visibility, circulation and standing. The present framework complements them by centering the catalogue's thematic organization and development over time.

A catalogue is a historical record shaped by editorial choices, series, collaborations and changing intellectual agendas. The ABYA-YALA catalogue combines broad long-term persistence with the largest post-prelude compositional shift in the most recent adjacent-window comparison. A cross-sectional count cannot capture both properties: similar totals may arise from continuous activity, episodic concentration or recent growth.

This longitudinal view connects publisher analysis to topic-evolution research but changes the unit of interpretation. Dynamic topic models and trajectory methods have mainly followed fields, venues, papers or authors [12,14,15]. Here, the trajectory records editorial development and recomposition rather than the evolution of a scientific field itself. Possible mechanisms include author supply, editorial strategy, co-publishing arrangements, institutional projects, documentary series and broader scientific change. Establishing causality would require evidence designed for those mechanisms.

5.2. Analytical value of multidimensional thematic profiles

The internal profile keeps volume, persistence, continuity, recent growth and within-family prominence separate, revealing configurations that an aggregate score would obscure. A high-volume line need not be continuous or recently growing, and a smaller line

may be prominent within its family. This design is consistent with responsible-evaluation principles emphasizing contextualized indicators and informed judgment [23], and with bibliodiversity research treating format, language, geography and infrastructure as separate dimensions [24].

The two-stage renewal analysis applies the same principle at line level. High recent growth can arise from a short episodic concentration, whereas a smaller growth ratio can satisfy continuity and recency conditions. The 36-scenario grid confirms that exact membership is parameter-dependent: Amazonian literature and culture is retained throughout, while the other reference lines are less stable. Future applications should therefore pre-specify renewal thresholds and test their sensitivity within each catalogue.

5.3. Documented expert curation as an inspectable analytical layer

Topic-model interpretation requires human evaluation because statistical coherence does not guarantee substantive interpretability [28,29]. The ABYA-YALA workflow records the algorithmic partition, curated map and document-level changes. Because expert input changes only a documented subset, it refines rather than replaces the machine-derived structure. Preserving both states makes the magnitude and consequences of that intervention quantifiable rather than leaving curation as an undocumented correction. This hybrid machine/expert process is therefore directly inspectable.

The same separation applies to LLM-assisted labels. GPT-4o proposed names from representative titles only after cluster membership was fixed, and expert review retained or renamed them. Membership, structural curation and labeling therefore have separate audit trails, an important distinction in LLM-supported scientometric workflows [37]. This separation also prevents a proposed label from being mistaken for evidence that the model changed document membership. The recorded transformations and supporting evidence are inspectable, but the study does not claim to explain the models' internal operation. The Kipu consolidation offers a concrete example: documentary-series records justified a structural correction, and catalogue registration dates constrain its chronological interpretation.

5.4. External scientific context for the semantic representation

OpenAlex, an open scholarly graph, supplies topic-country context for selected editorial lines across the ten pre-specified countries. The interpretation is comparative and topic-specific: the study uses topic and country attributes from this infrastructure to describe indexed activity around the mapped themes, whereas the circulation or citation of ABYA-YALA books would require publication-level linkage data such as citations, library holdings or translations.

This distinction is especially relevant in Latin American social sciences and humanities. OpenAlex can recover broader Spanish-language and SSH output than Scopus in some settings [31], but its language metadata and source coverage remain uneven [32,33]. The profiles therefore describe indexed activity within the selected database and countries. Their close agreement under equal-country and strategic weighting supports the qualitative ordering of focal lines within the ten-country comparison.

5.5. Documentary review refines screened research-front retention

Research Horizon addresses a different question from OpenAlex. Metadata similarity screens candidates, while corpus-wide documentary review classifies them as principal, complementary, specific, reserve or excluded. The retained-candidate fraction shows the denominator for every focal line and supports comparison within the screened set. This denominator is important because the number of reviewed Priority-1 candidates varies from two to six across lines.

The exclusion of some candidates with strong preliminary affinity shows the value of documentary review for assessing substantive relationships. Lexical or metadata proximity alone does not establish a meaningful research-front connection. The 11,325-to-73-to-30 audit trail records the selection process, including reserve and excluded cases, and the retained fractions describe this screened Priority-1 set.

5.6. Contribution of the cross-layer representation

Combining the sources shows how historical volume, recent renewal, indexed activity and research-front retention vary across editorial lines. Each focal line keeps its thematic identity, hierarchical context, longitudinal profile and links to external evidence. A large technical line, a smaller renewing educational line and a historically strong Indigenous-studies line therefore occupy different positions when all three sources are considered. These positions reveal configurations that no source can establish alone. Their comparison remains interpretable because the measures are not merged into a common score.

More generally, the sequence illustrates a pattern for knowledge-enhanced analytical systems. Heterogeneous texts become explicit topics, machine-learning output remains open to review, expert intervention is documented and structured external sources enrich the analysis without producing an opaque score. Each source continues to answer its original question rather than becoming a proxy for the others. Because every step remains traceable, the basis of cross-source interpretations can be inspected directly. This pattern may inform digital libraries, scientific data platforms, open knowledge systems and knowledge-enhanced web intelligence.

The framework is modular: the catalogue can be analyzed alone, and external sources or new indicators can be added when the research question and available metadata justify them. Their definitions, derivation and relation to the internal topics must remain explicit. This design supports cross-publisher replication while leaving case-specific thresholds and mappings open to pre-specification and validation.

5.7. Limitations

The framework was applied to a single publisher. Its analytical sequence and definitions are reusable, but the renewal thresholds, the nine focal lines, the macro-family cut, the ten-country OpenAlex set, the mapping roles and the Research Horizon screening strategy are case-specific choices whose transfer to other catalogues remains to be tested.

Two inputs to the semantic representation were not independently validated. SPECTER2 was developed for scientific documents across multiple fields [34] and is applied here to a heterogeneous book catalogue. The English translations used as model inputs were produced upstream and treated as fixed; the topic solution is reproducible from the frozen inputs, but its sensitivity to the translation step was not tested.

The reference map describes the stable density regions of the catalogue and leaves 1,580 of the 3,963 eligible records unassigned. The outlier scenarios show that global diversity is stable under reassignment, whereas the magnitude of recent compositional change and the exact renewal set depend on how diffuse records are treated: complete nearest-centroid forcing reduces the recent distance from 0.363 to 0.258 and expands the renewal set from four to ten, while the high-confidence scenario preserves the four reference renewals. The recent-recomposition finding therefore concerns the mapped stable core. Model granularity likewise depends on the clustering configuration, as the minimum-cluster-size perturbation in Appendix A.6 shows.

Curation, label revision, OpenAlex role assignment and the final Research Horizon classification rest on documented expert judgement. The decisions are preserved together

with their documentary basis, but they were not replicated by an independent coder, so inter-coder agreement could not be measured.

The external sources are bounded by design and coverage. The ten OpenAlex countries form a pre-specified comparison set rather than a global sample, and OpenAlex coverage and language metadata remain uneven across languages and fields [31–33]. Research Horizon is a licensed source, and its retention fractions describe the screened Priority-1 candidates.

The most recent catalogue years are incomplete. Records for 2025 and 2026 reflect the latest update rather than finalized annual production, which is why the complete 2022–2024 subwindow is used as a check; it yields a distance of 0.371 against 0.363 for the full window.

Finally, the renewal rule was formalized retrospectively from documented decisions. The 36-scenario grid shows that Amazonian literature and culture is retained in every combination while the other reference lines depend on the thresholds, which future applications should pre-specify.

6. Conclusions and future work

This article developed a multidimensional semantic framework for reconstructing a scholarly publisher's long-term thematic trajectory and relating it to external scientific evidence. The proof-of-concept combines the internal thematic map and its temporal development, OpenAlex indexed activity within a defined country set and retention of screened Research Horizon candidates after documentary review.

For the ABYA-YALA catalogue, the mapped stable semantic core combines long-term persistence with recent recomposition. Starting from the 70-cluster density-based partition, documented expert curation yields 68 analytical topics through a traceable document-level intervention. The final map has $HHI = 0.0259$, Hill diversity ${}^1D = 51.10$ and ${}^2D = 38.60$. The 2017–2021 to 2022–2026 composition differs substantially from the preceding state, and the direction of this shift persists when the recent window is restricted to complete 2022–2024 data and when high-confidence outliers are reassigned. Under the reference two-stage renewal specification, four lines qualify as recent renewal; the sensitivity grid identifies Amazonian literature and culture as the most stable member and shows boundary dependence for the remaining reference lines.

The external sources add complementary information. OpenAlex identifies line-specific geographic patterns of database-indexed scientific activity within the ten pre-specified countries. Research Horizon documentary review retains 23 of 30 Priority-1 candidates, with line-level fractions ranging from zero to complete retention. Together, the results show that historical volume, recent renewal, indexed activity and research-front retention can follow different patterns within the same editorial portfolio.

The transferable contribution is the analytical architecture: preserved algorithmic and curated states, an explicit audit trail for interpretation, separate longitudinal measures and documented links between the internal map and external scientific infrastructures. The case establishes feasibility and identifies renewal thresholds, focal-line definitions, country sets, hierarchy cuts and screening procedures as parameters for independent cross-publisher validation.

Future work should proceed through four concrete validation routes:

- Cross-publisher replication can test the stability and usefulness of the profile dimensions across university, commercial, disciplinary and multilingual presses.
- Independent double coding of topic labels, OpenAlex mappings and preserved Research Horizon corpora can quantify inter-rater agreement.
- Original-language title-and-abstract corpora can support translation audits and multilingual embedding comparisons.

- Persistent book identifiers can extend the external layer toward direct evidence of circulation or influence through citations, library holdings, translations or co-editions.

Overall, the study shows how a publisher catalogue can be treated as a longitudinal knowledge structure and compared with external evidence without conflating editorial development, indexed scientific activity and research-front relevance.

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Abbreviations

ARI	Adjusted Rand Index
c-TF-IDF	class-based Term Frequency–Inverse Document Frequency
EOM	Excess of Mass
HDBSCAN	Hierarchical Density-Based Spatial Clustering of Applications with Noise
HHI	Herfindahl–Hirschman Index
LLM	Large Language Model
NMI	Normalized Mutual Information
RQ	Research Question
SSH	Social Sciences and Humanities
UMAP	Uniform Manifold Approximation and Projection

Appendix A. Robustness and sensitivity details

Appendix A.1. Topic assignment across decades

Table A1 reports assignment among semantically eligible records. The absence of an abrupt historical collapse does not establish that outliers are missing at random, but it shows that the 39.9% overall noise share is not generated by one isolated period.

Table A1. HDBSCAN assignment rate among semantically eligible books in each decade of the catalogue

Period	Eligible	Assigned	Assignment rate
1978–1989	164	107	65.2%
1990–1999	781	500	64.0%
2000–2009	954	570	59.7%
2010–2019	1,515	881	58.2%
2020–2026	549	325	59.2%

Appendix A.2. Hierarchical family cut

The number of macro-families changes monotonically around the selected hierarchy threshold, with a short plateau at nine families (Table A2). This is why the family layer is interpreted as a useful mesoscale organization rather than a uniquely identified taxonomy.

Appendix A.3. Outlier-reassignment stress tests

The reference solution leaves HDBSCAN noise unassigned. Table A3 compares that state with two deliberately different reassignment scenarios. The high-confidence scenario

Table A2. Sensitivity of the number of macro-families to the hierarchical cut threshold

Cut threshold	4.00	4.25	4.50	4.75	5.00	5.25	5.50
Macro-families	12	11	10	9	9	8	7

assigns 166 noise records whose centroid similarity and assignment margin both exceed 25th-percentile thresholds derived from assigned records; complete forcing assigns all 1,580 noise records to the nearest final-topic cosine centroid. The controlled scenario leaves global diversity and the four reference renewals unchanged. Complete forcing retains those four lines but adds six further renewals, while also reducing the recent compositional distance. This contrast shows that line-level temporal classification is more sensitive than overall diversity to aggressive outlier treatment.

Table A3. Sensitivity of global diversity, recent compositional change and renewal membership to outlier reassignment

Scenario	Mapped	HHI	^1D	^2D	Recent distance	Renewals
Reference stable core	2,383	0.0259	51.10	38.60	0.363	4
High-confidence reassignment	2,549	0.0256	51.34	39.08	0.334	4
All noise forced to nearest centroid	3,963	0.0242	53.01	41.33	0.258	10

In the complete-forcing scenario, all four reference renewals remain selected and six additional topics enter the renewal set. The robustness result is therefore asymmetric: the reference lines are not lost, but the specificity of the four-line classification depends on preserving diffuse observations as noise rather than forcing complete coverage.

Appendix A.4. Renewal-threshold perturbation

The reference two-stage renewal specification is evaluated for threshold sensitivity. Across 36 combinations of growth-ratio, episodicity and final-year thresholds, the number of retained renewals varies from one to seven. Table A4 reports selection frequencies for every topic retained in at least one perturbation. Amazonian literature and culture is the only line retained under all tested combinations.

Table A4. Selection frequency of each topic under the 36 renewal-threshold combinations

Topic	Share of scenarios retained
Amazonian literature and culture	100.0%
Inclusive educational innovation	75.0%
Gender rights and violence	66.7%
Intercultural bilingual education	66.7%
Interculturality and cultural educational management	33.3%
Anthropology and material relations	33.3%
Digital media and social networks	25.0%

Appendix A.5. Recent-window completeness

The largest recent reference distance is not inflated by the incomplete 2025–2026 tail. The distance from 2017–2021 to 2022–2026 is 0.363, whereas replacing the second period with complete 2022–2024 data yields 0.371. Under the high-confidence outlier scenario the corresponding values are 0.334 and 0.341. The recent recomposition signal therefore persists when incomplete end years are excluded.

Appendix A.6. Clustering-granularity diagnostics

A cross-implementation HDBSCAN perturbation was run on the archived five-dimensional UMAP representation. The reference-like setting (`min_cluster_size` = 10 with the adjusted scikit-learn `min_samples` semantics) closely matches the frozen initial algorithmic solution (ARI 0.978; NMI 0.985). Moving the minimum cluster size from 8 to 12 changes the number of clusters from 92 to 55 and the noise share from 39.7% to 45.2% (Table A5). These values demonstrate expected granularity dependence rather than a uniquely determined cluster count.

Table A5. Cross-implementation sensitivity of the HDBSCAN partition to the minimum cluster size

Minimum cluster size	Clusters	Noise share	ARI vs. frozen	NMI vs. frozen
8	92	0.397	0.706	0.851
9	84	0.414	0.767	0.881
10	70	0.402	0.978	0.985
11	62	0.424	0.837	0.903
12	55	0.452	0.740	0.855

An exact historical UMAP `n_neighbors` rerun was not reconstructed from the archived runtime. As a narrower diagnostic, the mean share of same-topic neighbors in the original 768-dimensional SPECTER2 space decreases from 0.748 at $k = 10$ to 0.673 at $k = 15$, 0.614 at $k = 20$ and 0.528 at $k = 30$. This indicates that the final topics retain substantial local semantic coherence near the reference neighborhood scale, but it should not be read as a direct UMAP-parameter sensitivity experiment.

Appendix A.7. Characterization of unassigned records and duplicate bound

Within the preserved analytical state, unassigned records have a median original-title length of 66 characters compared with 60 for assigned records; median model-text lengths are 64 and 59, respectively. Exact duplicate-title shares are 12.9% and 12.5%. These diagnostics do not support a simple "short text produces noise" explanation. Complete original-language and series metadata are not preserved at sufficient resolution to test all other mechanisms that could plausibly contribute to non-assignment.

At least three residual duplicate pairs are known after upstream cleaning. If each pair contributes one excess record, the maximum known global inflation is three records, or 0.126% of the 2,383 mapped books. An extreme local upper bound occurs if all three excess records belong to a smallest ten-book topic, in which case that topic's count could be inflated by 30%. This worst case is not observed evidence about the duplicate locations; it is reported only to bound the possible effect of the known minimum duplicate count.

Appendix A.8. OpenAlex territorial scenario

A pre-specified strategic territorial scenario assigns 60% of the line-level weight to Ecuador, Mexico and Spain and 40% to the other seven countries. The equal-country mean serves as the scientific reference, while the 60/40 weighting is retained as an alternative strategic scenario. Table A6 compares the equal-country mean with the strategic scenario. The ordering is highly similar ($\rho = 0.983$), but the differences are retained to prevent the strategic weighting from being mistaken for an objective ranking.

Appendix A.9. OpenAlex weight perturbation

To assess whether the geographic pattern depends on the exact component and mapping-role weights, we independently perturbed the five topic-country component weights and the four mapping-role weights by ±25% in 5,000 Monte Carlo simulations. Each simulation was renormalized to the same 0–100 line-country scale. The aggregate

Table A6. OpenAlex line summaries under the equal-country reference and the strategic 60/40 scenario

Line	Equal-country mean	60/40 scenario	Difference
Indigenous movement and plurinationality	45.26	46.45	+1.19
Amazonian literature and culture	42.81	43.00	+0.20
Inclusive educational innovation	39.36	42.57	+3.21
Gender rights and violence	39.02	39.73	+0.71
Education and Indigenous autonomy	37.99	38.96	+0.98
Intercultural bilingual education	36.31	36.32	+0.01
Engineering and technology	36.30	36.79	+0.49
Anthropology and archaeology	35.26	34.89	-0.36
Histories, migration and speculative narratives	32.90	33.07	+0.18

country ordering was highly stable (Table A7): the United States ranked first in all simulations, while Spain and Brazil remained in the top three in all simulations. At line level, the leading country was unchanged in all simulations for six lines and in at least 99.5% for two additional lines. Amazonian literature and culture was the most weight-sensitive case: Brazil remained the leading country in 70.0% of simulations, followed by the United States (17.2%) and Spain (12.8%). This sensitivity is reported as a property of that line rather than suppressed by the aggregate ranking.

Table A7. Monte Carlo stability of selected OpenAlex geographic conclusions (5,000 simulations)

Quantity	Modal outcome	Stability
Aggregate country rank 1	United States	100.0%
Aggregate countries in top 3	United States, Spain, Brazil	100.0% each
Movement/plurinationality line leader	Argentina	99.6%
Anthropology/archaeology line leader	United States	99.7%
Amazonian literature/culture line leader	Brazil	70.0%

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